

Basic tool: orientation tests

Rodrigo Silveira

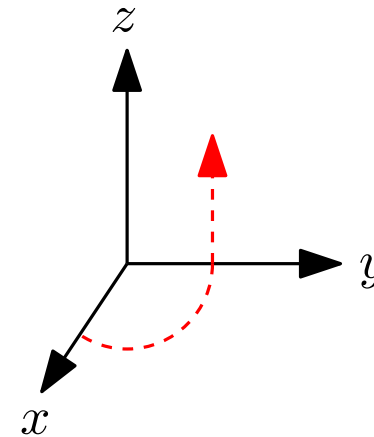
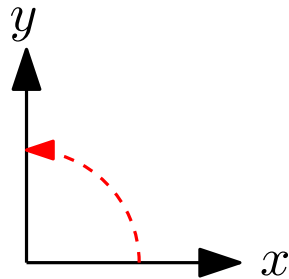
Based on slides by **Vera Sacristán**

Computational Geometry
Facultat d'Informàtica de Barcelona
Universitat Politècnica de Catalunya

BASIC TOOL: ORIENTATION TESTS

WARNING

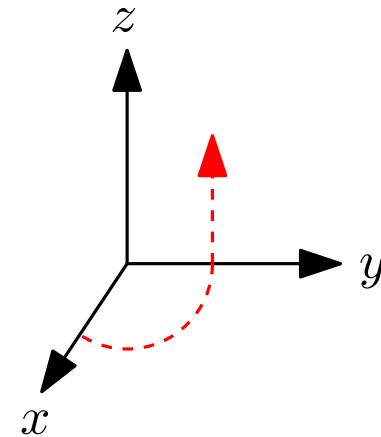
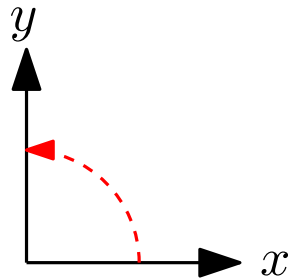
Throughout this entire course, as in all textbooks, it is assumed that both the plane and the 3-dimensional space are positively oriented.



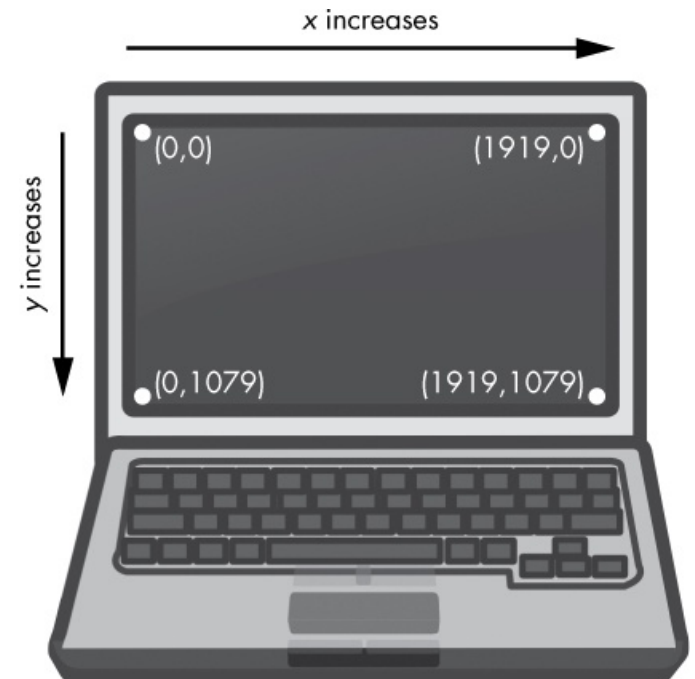
BASIC TOOL: ORIENTATION TESTS

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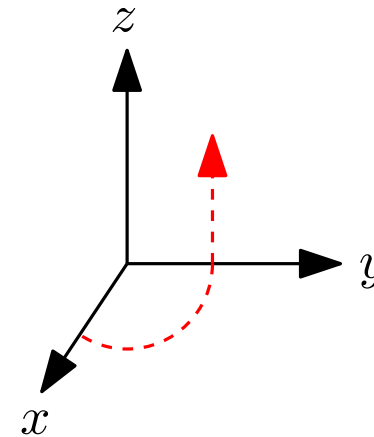
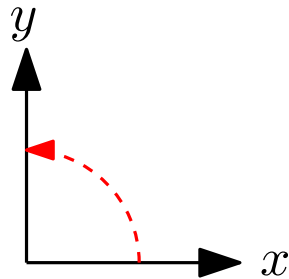
Please be aware that most graphics systems orient the screen of your computers and the 3-dimensional space negatively!



BASIC TOOL: ORIENTATION TESTS

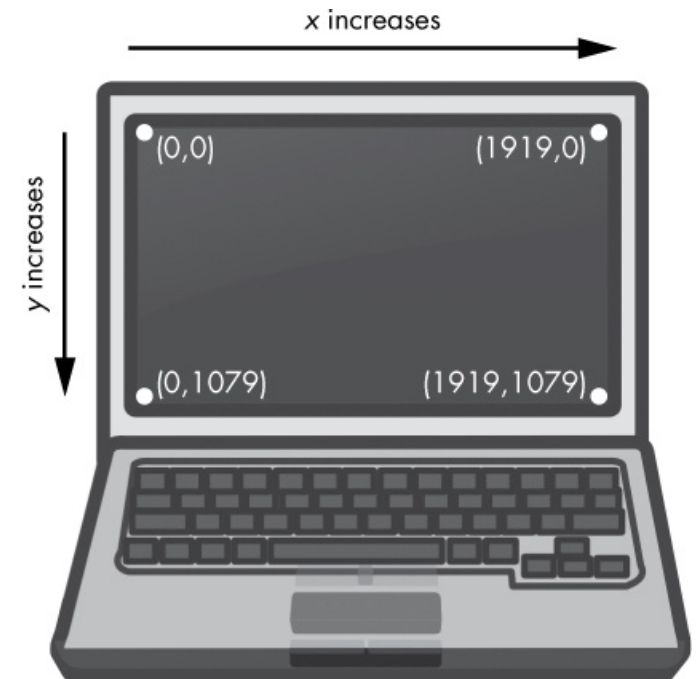
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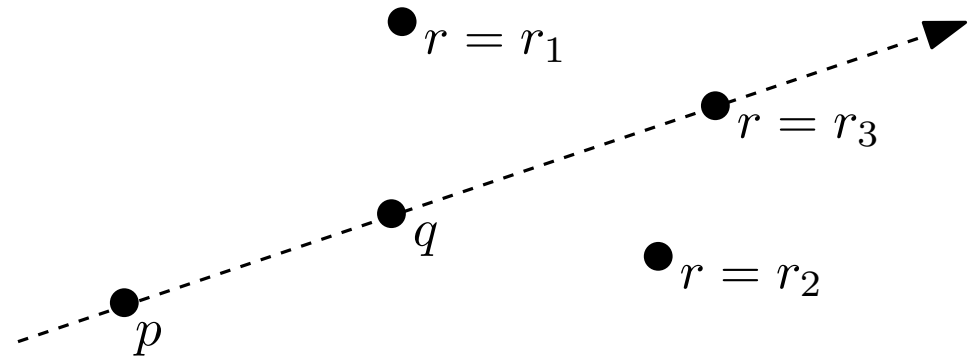
Keep this in mind for the lab!



BASIC TOOL: ORIENTATION TESTS

Orientation test in $\mathbb{R}^2$

Given 3 points p, q, r in the plane, efficiently and robustly decide whether r lies to the left, to the right or on the oriented line pq .

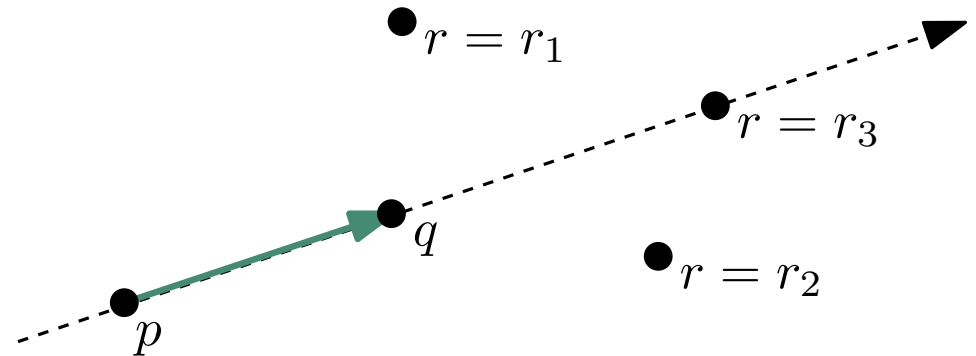


BASIC TOOL: ORIENTATION TESTS

Orientation test in $\mathbb{R}^2$

Given 3 points p, q, r in the plane, efficiently and robustly decide whether r lies to the left, to the right or on the oriented line pq .

Consider the vectors $\overrightarrow{pq}$...

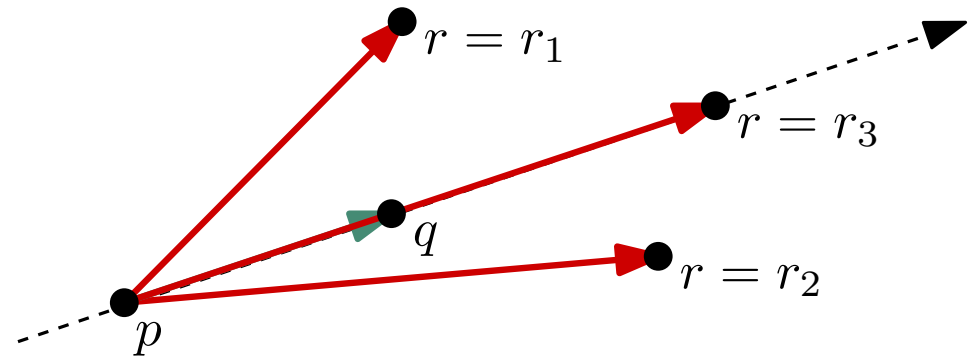


BASIC TOOL: ORIENTATION TESTS

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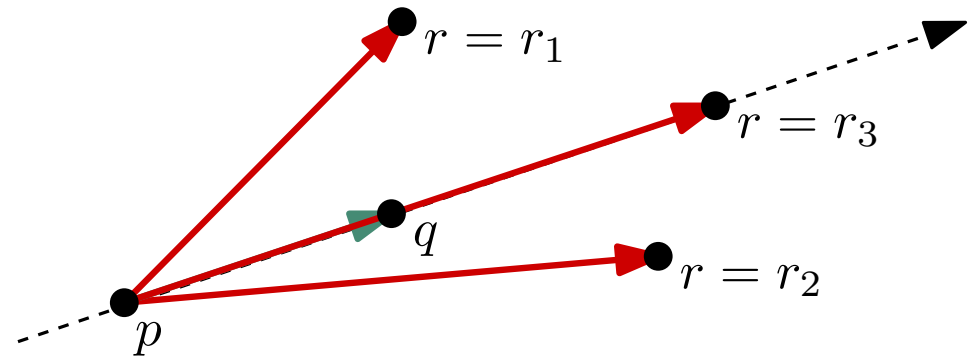


BASIC TOOL: ORIENTATION TESTS

Orientation test in $\mathbb{R}^2$

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Consider the vectors $\overrightarrow{pq}$ and $\overrightarrow{pr}$.



Point r lies on the line $pq \iff \begin{vmatrix} q_x - p_x & r_x - p_x \\ q_y - p_y & r_y - p_y \end{vmatrix} = 0$

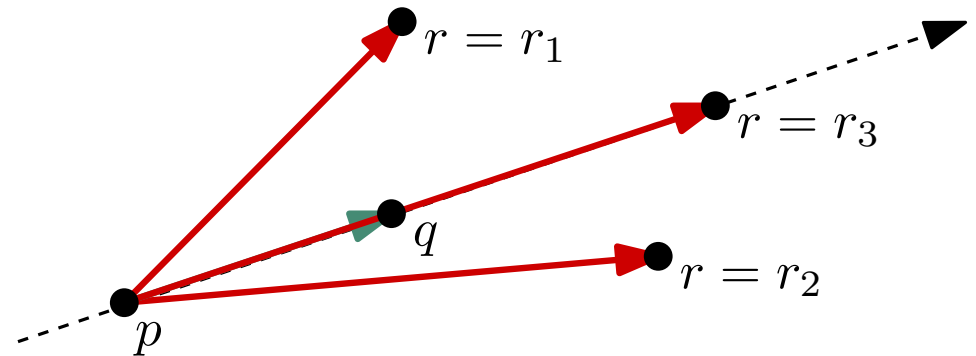
Recall what the determinant of a 2×2 matrix is!

BASIC TOOL: ORIENTATION TESTS

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Point r lies on the line $pq \iff \begin{vmatrix} q_x - p_x & r_x - p_x \\ q_y - p_y & r_y - p_y \end{vmatrix} = 0$ Recall what the determinant of a 2×2 matrix is!

Point r lies to the left of the oriented line $pq \iff \begin{vmatrix} q_x - p_x & r_x - p_x \\ q_y - p_y & r_y - p_y \end{vmatrix} > 0$

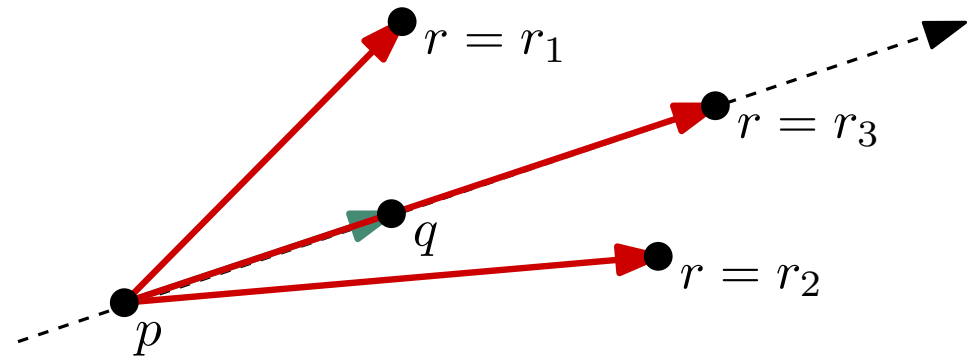
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BASIC TOOL: ORIENTATION TESTS

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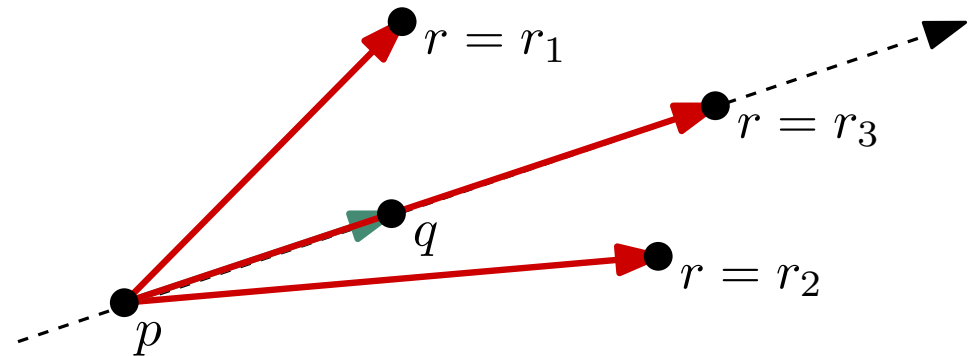
Try it in your favorite application! For instance, in [desmos.com](https://www.desmos.com)

BASIC TOOL: ORIENTATION TESTS

Orientation test in $\mathbb{R}^2$

Given 3 points p, q, r in the plane, efficiently and robustly decide whether r lies to the left, to the right or on the oriented line pq .

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Notation

$$\det(p, q, r) = \begin{vmatrix} q_x - p_x & r_x - p_x \\ q_y - p_y & r_y - p_y \end{vmatrix} = \begin{vmatrix} p_x & q_x & r_x \\ p_y & q_y & r_y \\ 1 & 1 & 1 \end{vmatrix}$$

BASIC TOOL: ORIENTATION TESTS

Relative position point - line

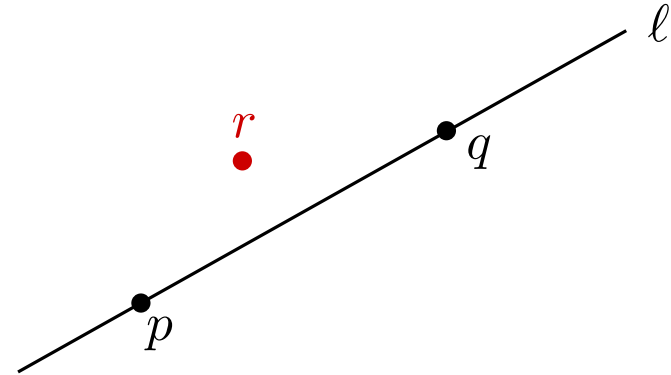
Input:

ℓ : a line (through points p and q)

r : a point

Output:

Relative position of r w.r.t. ℓ .



BASIC TOOL: ORIENTATION TESTS

Relative position point - line

Input:

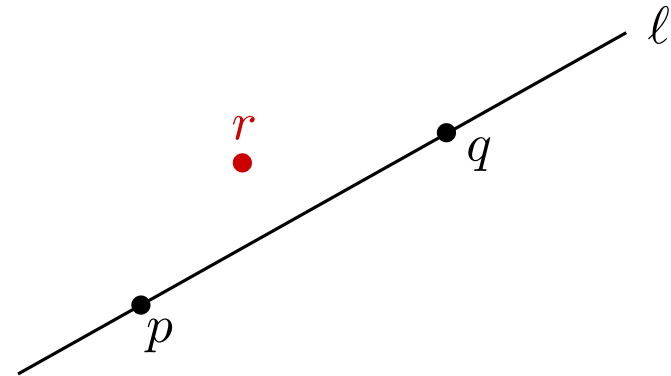
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Output:

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Equivalent question: do p, q, r turn left or right?



BASIC TOOL: ORIENTATION TESTS

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Input:

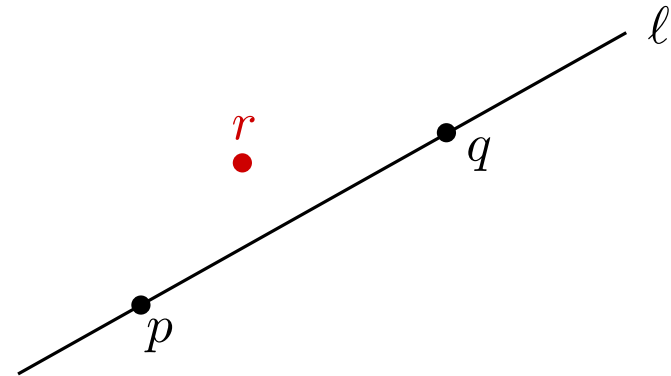
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Intersection test 1: line segment - line

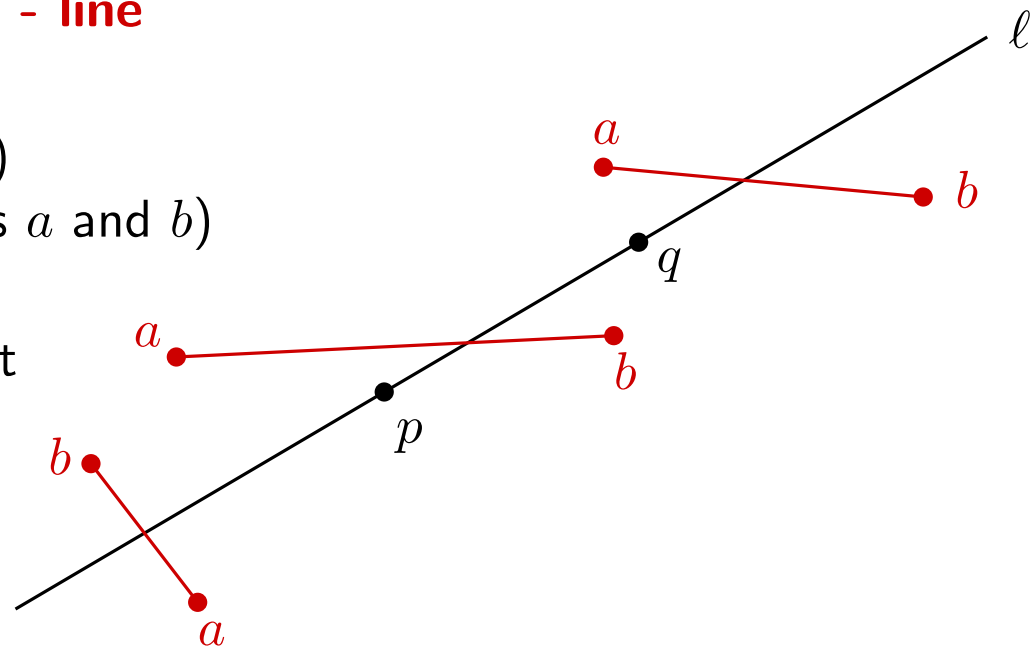
Input:

ℓ : a line (through points p and q)

s : a line segment (with endpoints a and b)

Output:

Yes, they intersect/No, they don't



BASIC TOOL: ORIENTATION TESTS

Relative position point - line

Input:

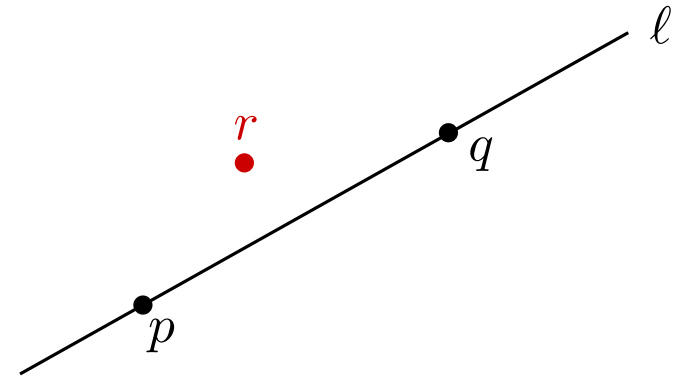
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Intersection test 1: line segment - line

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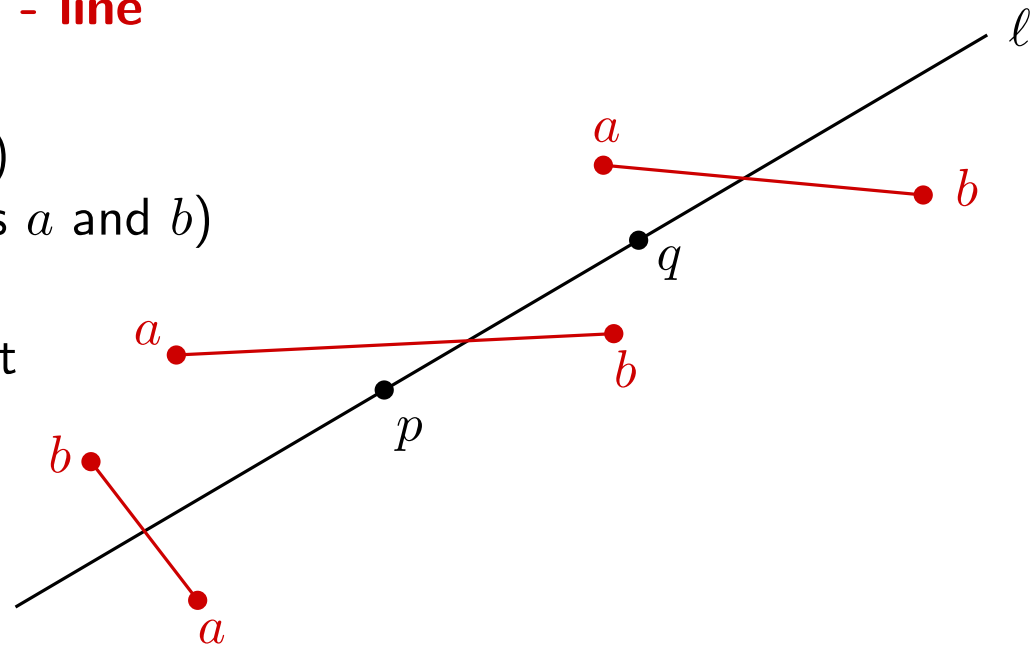
ℓ : a line (through points p and q)

s : a line segment (with endpoints a and b)

Output:

Yes, they intersect/No, they don't

Equivalent question: are a and b on opposite sides of ℓ ?



BASIC TOOL: ORIENTATION TESTS

Intersection test 2: line segment - halfline

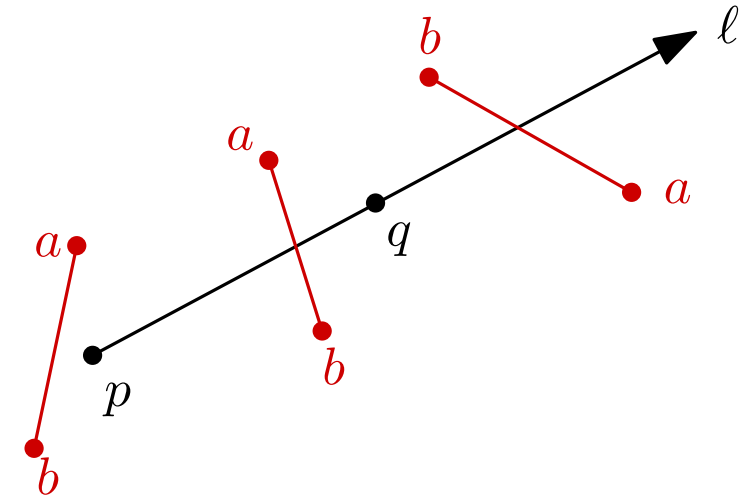
Input:

ℓ : halfline (from p through q)

s : a line segment (with endpoints a and b)

Output:

Yes, they intersect/No, they don't



BASIC TOOL: ORIENTATION TESTS

Intersection test 2: line segment - halfline

Input:

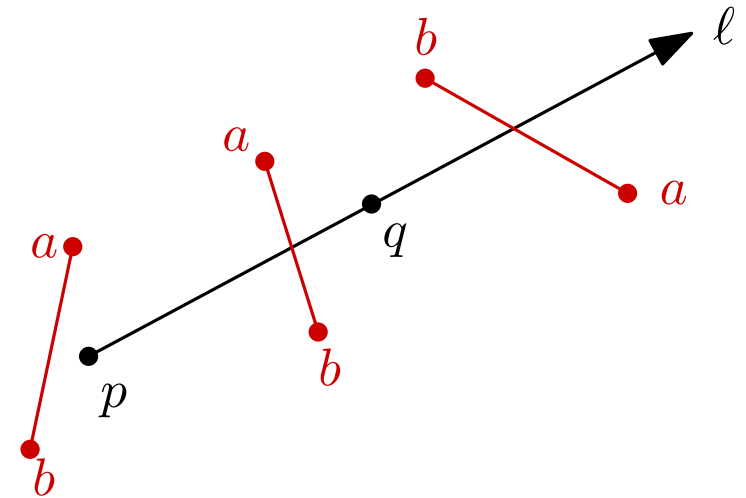
ℓ : halfline (from p through q)

s : a line segment (with endpoints a and b)

Output:

Yes, they intersect/No, they don't

Equivalent questions: 1) Are a and b on opposite sides of the line through ℓ ? 2) If so, assuming a left of ℓ , do a, b, p make a right turn?



BASIC TOOL: ORIENTATION TESTS

Intersection test 2: line segment - halfline

Input:

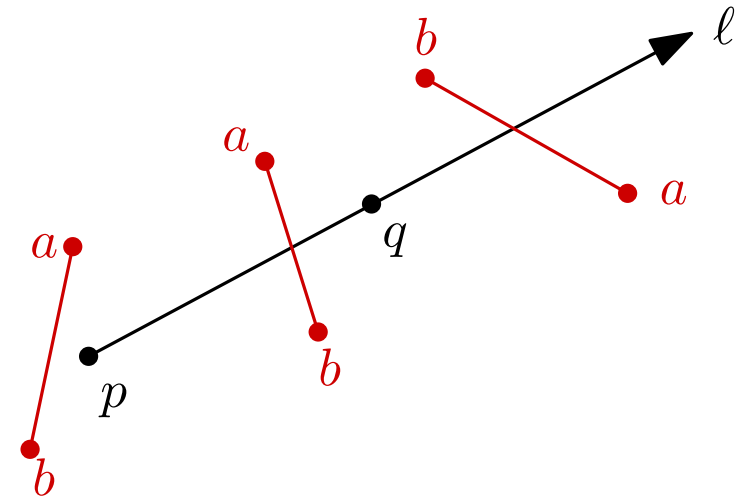
ℓ : halfline (from p through q)

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Output:

Yes, they intersect/No, they don't

Equivalent questions: 1) Are a and b on opposite sides of the line through ℓ ? 2) If so, assuming a left of ℓ , do a, b, p make a right turn?



Intersection test 3: line segment - line segment

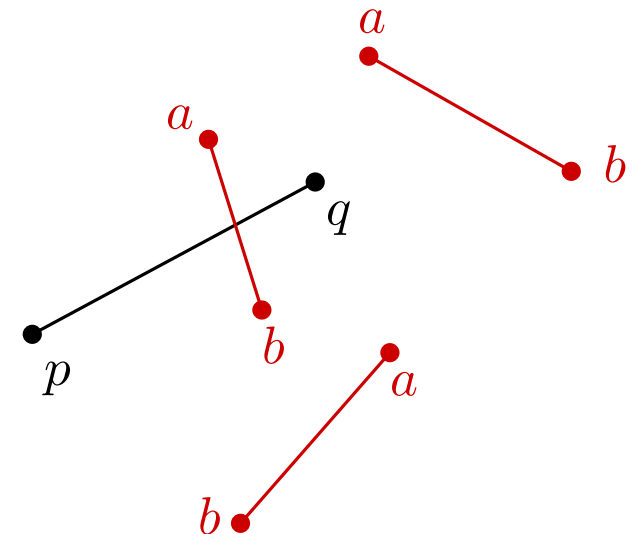
Input:

s_1 : a line segment (with endpoints p and q)

s_2 : a line segment (with endpoints a and b)

Output:

Yes, they intersect/No, they don't



BASIC TOOL: ORIENTATION TESTS

Intersection test 2: line segment - halfline

Input:

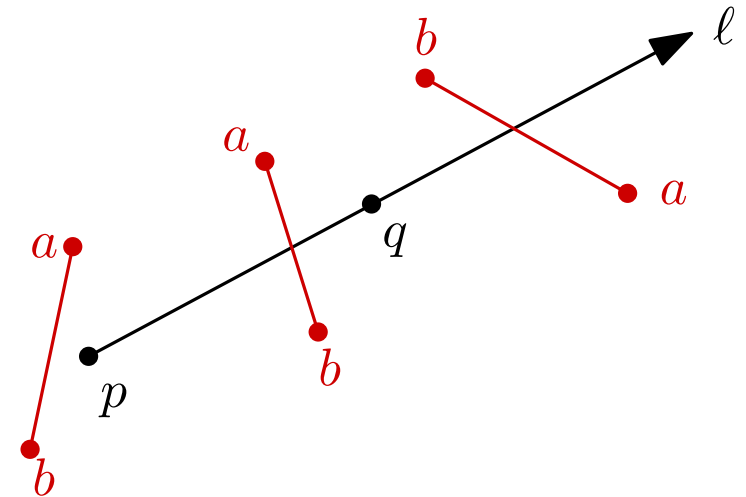
ℓ : halfline (from p through q)

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Output:

Yes, they intersect/No, they don't

Equivalent questions: 1) Are a and b on opposite sides of the line through ℓ ? 2) If so, assuming a left of ℓ , do a, b, p make a right turn?



Intersection test 3: line segment - line segment

Input:

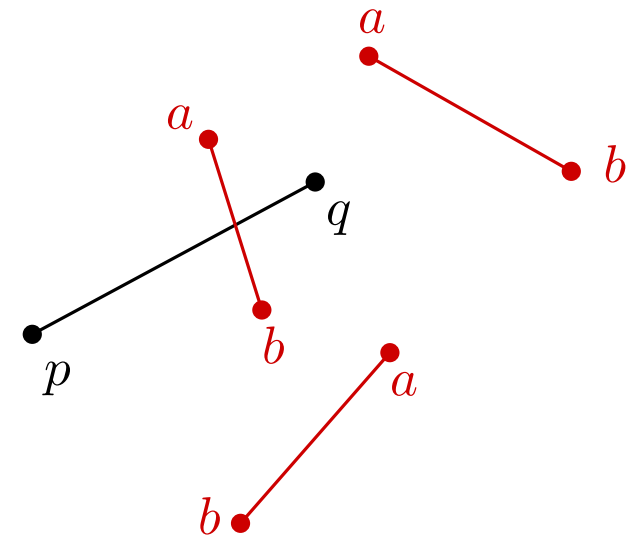
s_1 : a line segment (with endpoints p and q)

s_2 : a line segment (with endpoints a and b)

Output:

Yes, they intersect/No, they don't

Equivalent questions: 1) Are a, b on opposite sides of the line through p, q AND 2) Are p, q on opposite sides of the line through a, b ?



BASIC TOOL: ORIENTATION TESTS

Intersection test 2: line segment - halfline

Input:

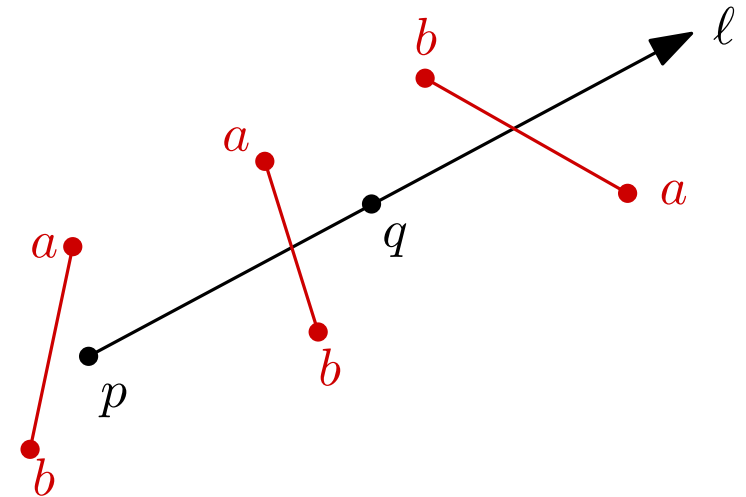
ℓ : halfline (from p through q)

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Output:

Yes, they intersect/No, they don't

Equivalent questions: 1) Are a and b on opposite sides of the line through ℓ ? 2) If so, assuming a left of ℓ , do a, b, p make a right turn?



Topic of lab 1

Intersection test 3: line segment - line segment

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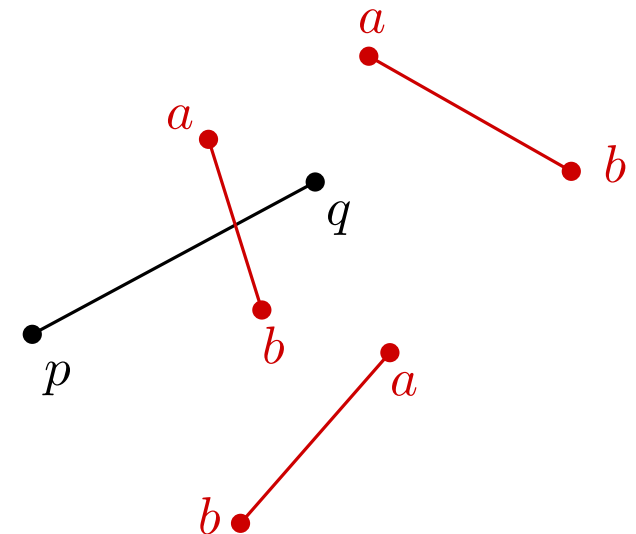
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Output:

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Equivalent questions: 1) Are a, b on opposite sides of the line through p, q AND 2) Are p, q on opposite sides of the line through a, b ?



SOME OTHER USES

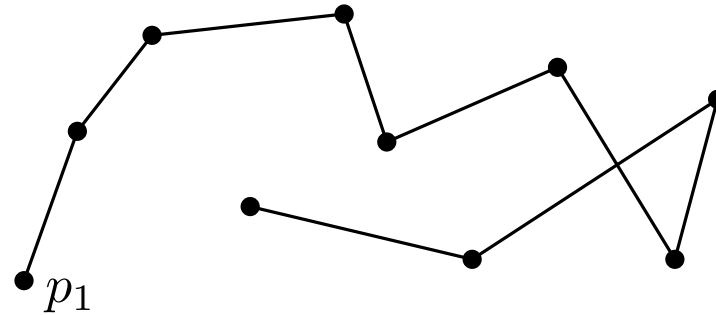
Turn orientation

Input:

A polygonal line $p_1, p_2, \dots, p_n$

Output:

Left/right classification of its turns



BASIC TOOL: ORIENTATION TESTS

SOME OTHER USES

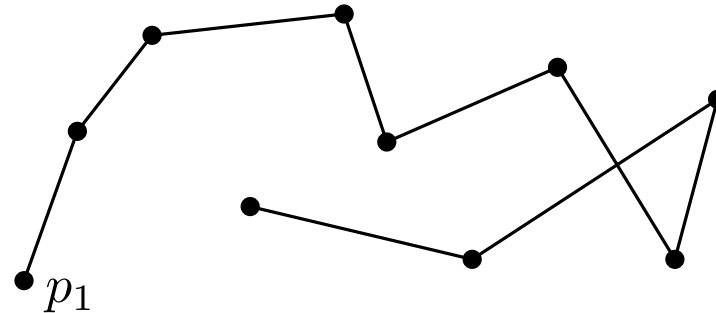
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Output:

Left/right classification of its turns



Point in triangle test

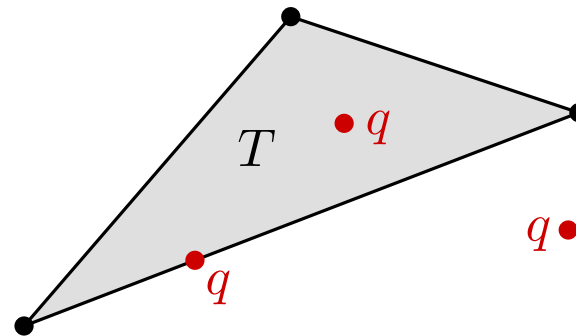
Input:

A triangle T with vertices p_1, p_2, p_3

A query point q

Output:

Relative position of q w.r.t. T



BASIC TOOL: ORIENTATION TESTS

SOME OTHER USES

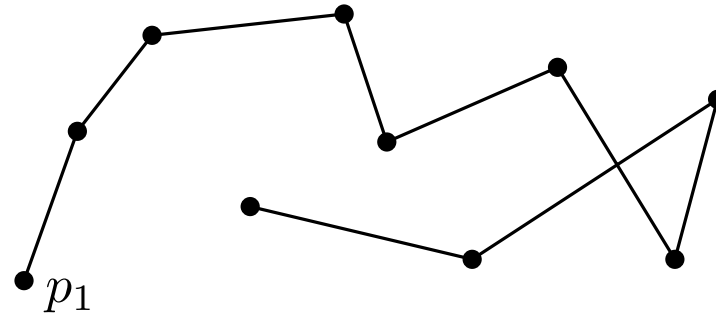
Turn orientation

Input:

A polygonal line $p_1, p_2, \dots, p_n$

Output:

Left/right classification of its turns



Topic of lab 2

Point in triangle test

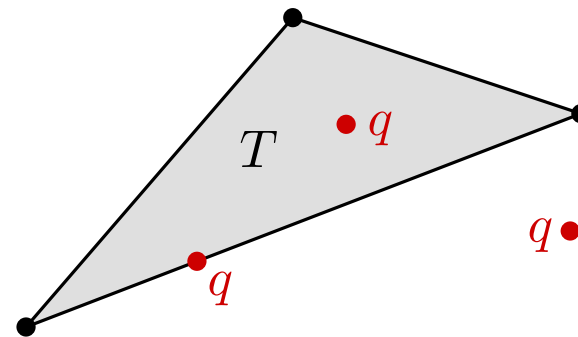
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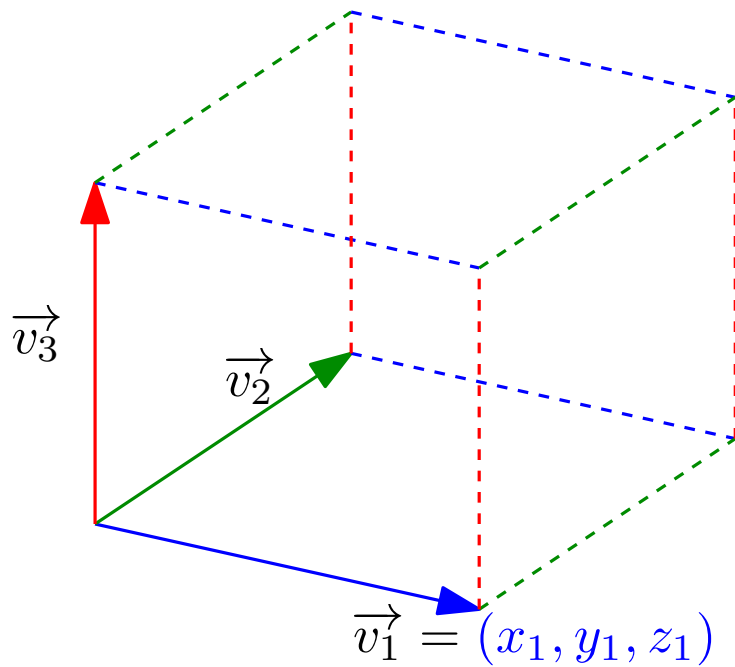
Output:

Relative position of q w.r.t. T



What happens in $\mathbb{R}^3$?

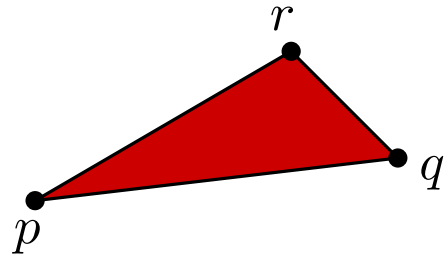
What happens in $\mathbb{R}^3$?



Volume of parallelepiped =
$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix}$$

BASIC TOOL: ORIENTATION TESTS

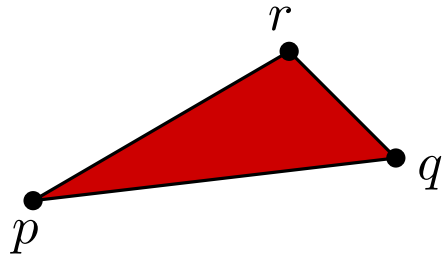
Oriented area of a triangle



$$\text{Oriented area } (p, q, r) = \frac{1}{2} \begin{vmatrix} p_x & q_x & r_x \\ p_y & q_y & r_y \\ 1 & 1 & 1 \end{vmatrix}$$

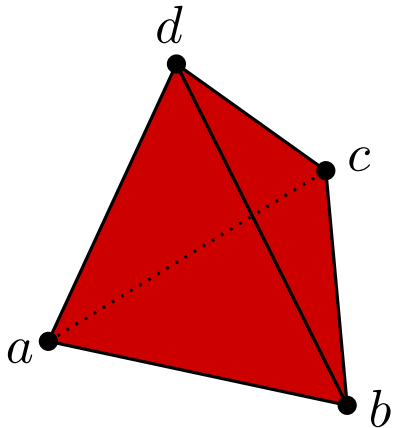
BASIC TOOL: ORIENTATION TESTS

Oriented area of a triangle



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Oriented volume of a tetrahedron

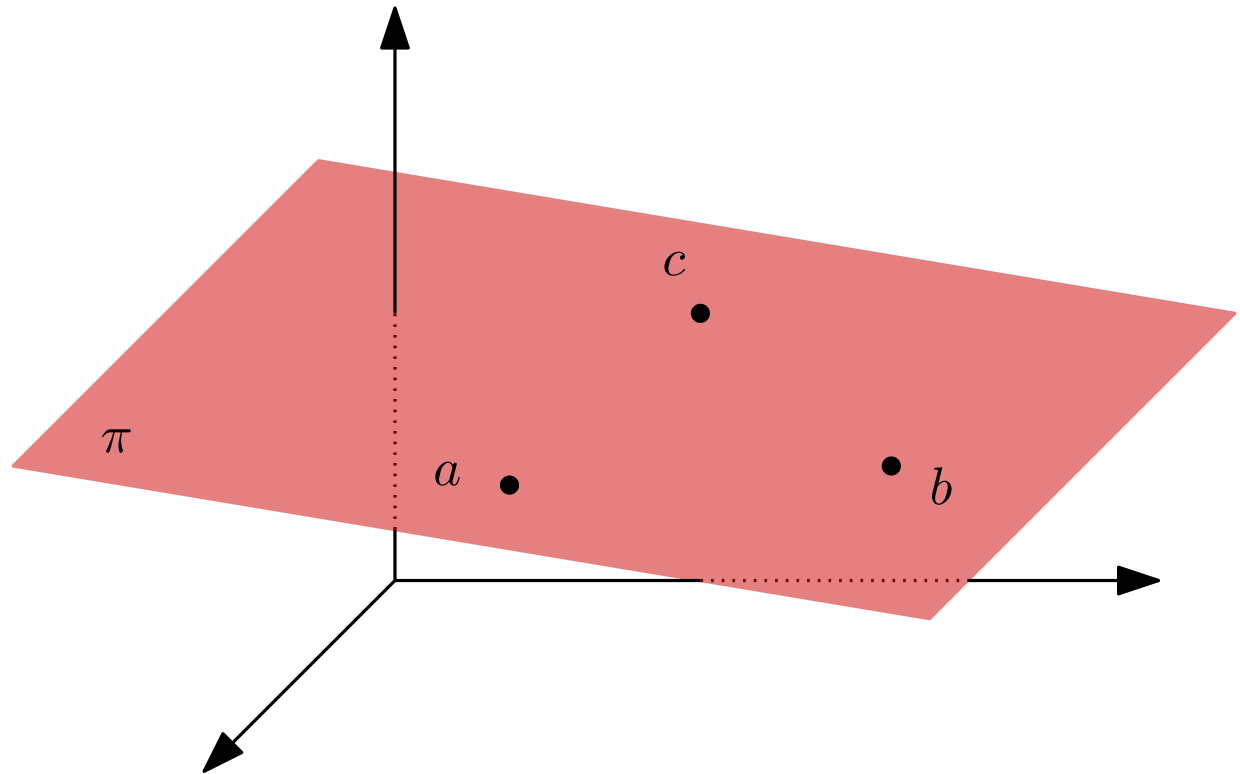


$$\text{Oriented volume } (a, b, c, d) = \frac{1}{6} \begin{vmatrix} a_x & b_x & c_x & d_x \\ a_y & b_y & c_y & d_y \\ a_z & b_z & c_z & d_z \\ 1 & 1 & 1 & 1 \end{vmatrix}$$

BASIC TOOL: ORIENTATION TESTS

Relative position point-plane

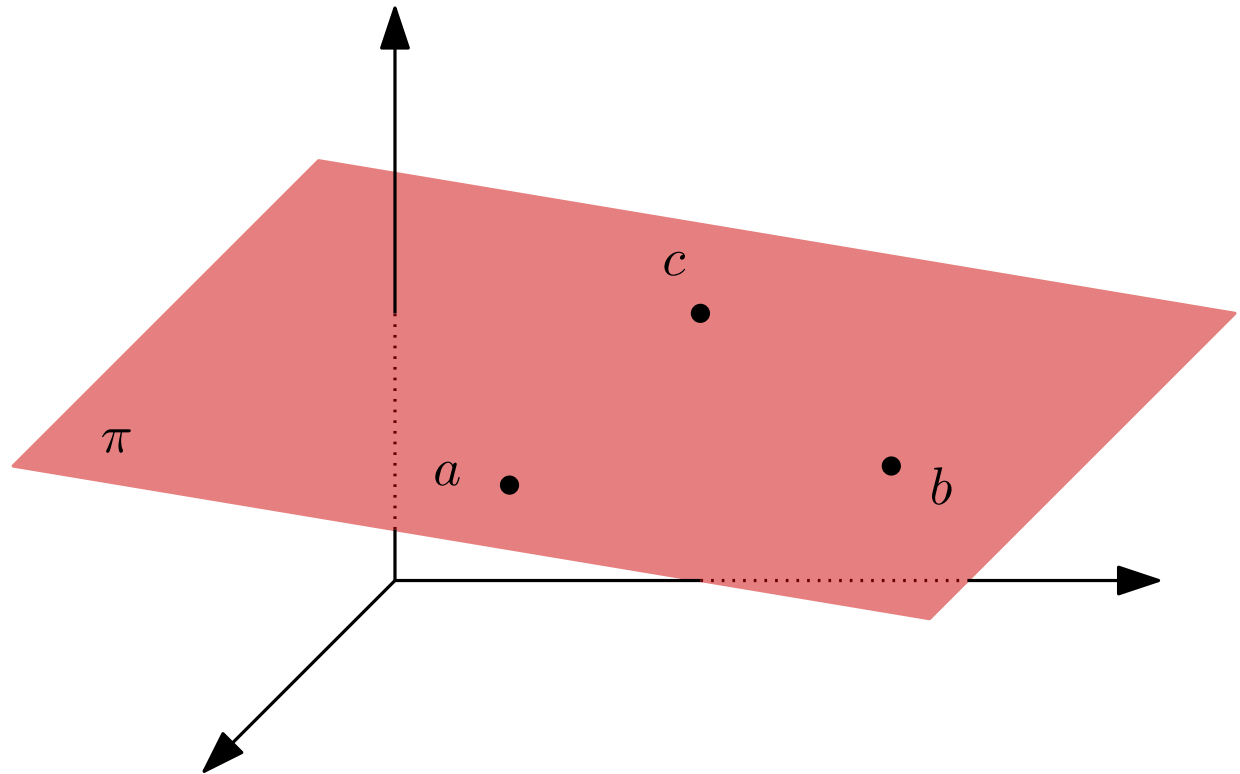
If a , b , and c are not aligned, they define a plane π , and two associated halfspaces π^+ and π^- .



BASIC TOOL: ORIENTATION TESTS

Relative position point-plane

If a , b , and c are not aligned, they define a plane π , and two associated halfspaces π^+ and π^- .



Point x lies in the halfspace $\pi^+ \iff \det(x, a, b, c) > 0$.

Point x lies in the plane $\pi \iff \det(x, a, b, c) = 0$.

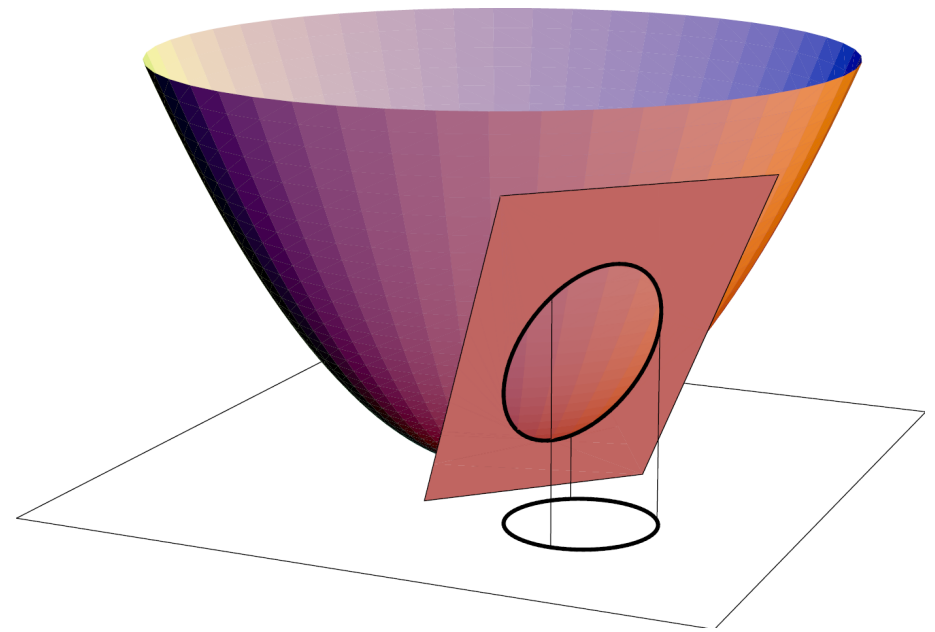
Point x lies in the halfspace $\pi^- \iff \det(x, a, b, c) < 0$.

BASIC TOOL: ORIENTATION TESTS

2D application: Relative position point-circle

Proposition 1

The intersection of the paraboloid whose equation is $z = x^2 + y^2$ with a non-vertical plane is a curve that projects orthogonally onto a circle in the plane $z = 0$.



2D application: Relative position point-circle

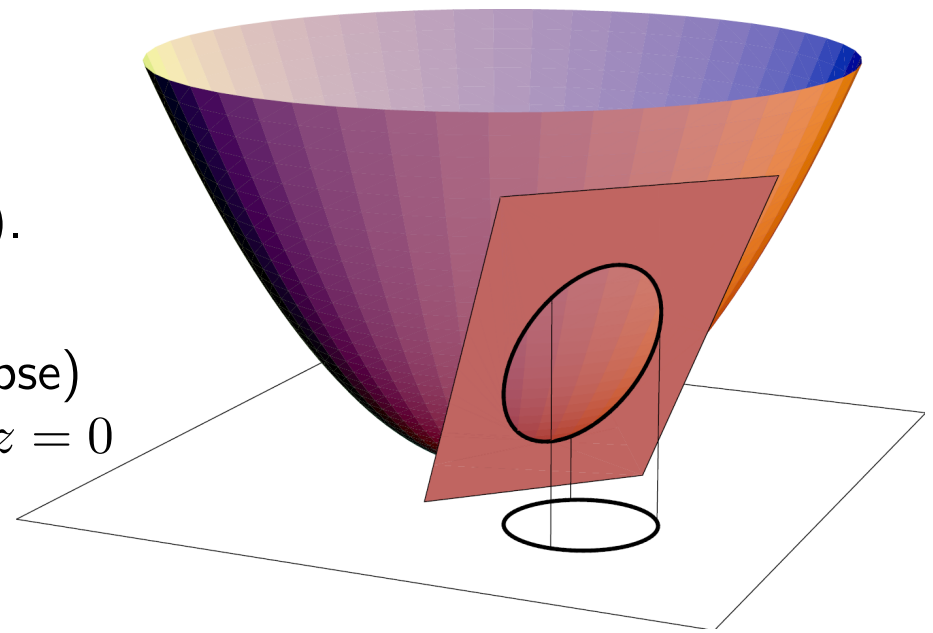
Proposition 1

The intersection of the paraboloid whose equation is $z = x^2 + y^2$ with a non-vertical plane is a curve that projects orthogonally onto a circle in the plane $z = 0$.

Proof:

$$\left. \begin{array}{l} z = x^2 + y^2 \\ z = 2ax + 2by + c \end{array} \right\} \quad \left. \begin{array}{l} x^2 + y^2 = 2ax + 2by + c \\ (x - a)^2 + (y - b)^2 = c + a^2 + b^2 \end{array} \right\}$$

- If $c + a^2 + b^2 < 0$,
the intersection is empty.
- If $c + a^2 + b^2 = 0$,
the intersection is the point $(a, b, a^2 + b^2)$.
- If $c + a^2 + b^2 > 0$,
the intersection is a curve (in fact, an ellipse)
that projects onto the circle of the plane $z = 0$
whose center is (a, b) and whose radius is
 $r = \sqrt{c + a^2 + b^2}$.



BASIC TOOL: ORIENTATION TESTS

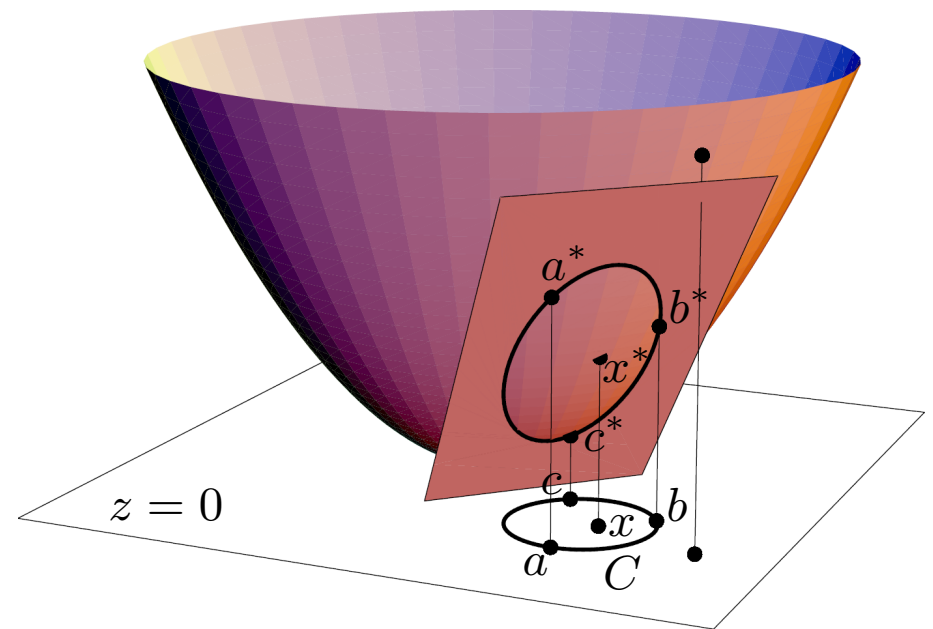
2D application: Relative position point-circle

Proposition 2

Let x, a, b, c be four points in the plane $z = 0$, and let x^*, a^*, b^*, c^* respectively be their vertical projections onto the paraboloid $z = x^2 + y^2$. If a, b, c are not aligned, let C be the circle through a, b, c , and let π be the plane through a^*, b^*, c^* .

Then:

- The point x lies in the circle C if and only if x^* lies in the plane π .
- The point x lies in the interior of the circle C if and only if x^* lies in the lower half-space determined by π .
- The point x lies in the exterior of the circle C if and only if x^* lies in the upper half-space determined by π .



BASIC TOOL: ORIENTATION TESTS

2D application: Relative position point-circle

Proposition 2

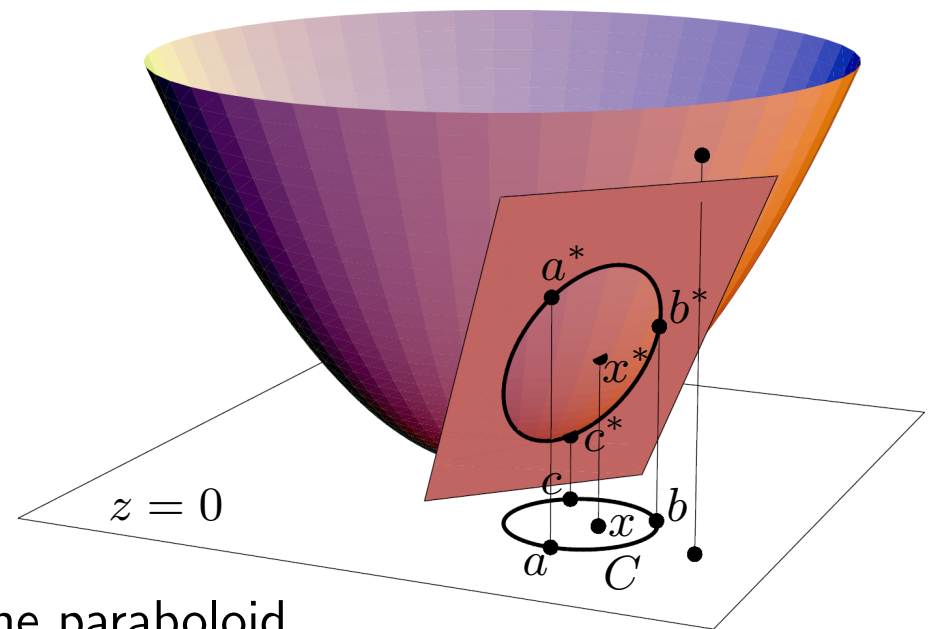
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Proof:

Due to Proposition 1 and the convexity of the paraboloid.



2D application: Relative position point-circle

Corollary

Let a , b and c be three non aligned points in the plane, that appear angularly sorted in counterclockwise order in the circle C that they determine.

Let x be any point in the plane.

Then:

- The point x lies in the circle C if and only if $\det(x^*, a^*, b^*, c^*) = 0$.
- The point x lies in the interior of C if and only if $\det(x^*, a^*, b^*, c^*) < 0$.
- The point x lies in the exterior of C if and only if $\det(x^*, a^*, b^*, c^*) > 0$.

2D application: Relative position point-circle

Observation

In order to compute the determinant of the previous corollary, it is convenient to do the calculations in terms of the differences between the values of the coordinates of the points involved, and to avoid making calculations (specially, products) in terms of the coordinate values, if possible:

$$\begin{vmatrix} x_1 & x_2 & x_1^2 + x_2^2 & 1 \\ a_1 & a_2 & a_1^2 + a_2^2 & 1 \\ b_1 & b_2 & b_1^2 + b_2^2 & 1 \\ c_1 & c_2 & c_1^2 + c_2^2 & 1 \end{vmatrix} = \begin{vmatrix} b_1 - a_1 & b_2 - a_2 & (b_1 - a_1)(b_1 + a_1) + (b_2 - a_2)(b_2 + a_2) \\ c_1 - a_1 & c_2 - a_2 & (c_1 - a_1)(c_1 + a_1) + (c_2 - a_2)(c_2 + a_2) \\ x_1 - a_1 & x_2 - a_2 & (x_1 - a_1)(x_1 + a_1) + (x_2 - a_2)(x_2 + a_2) \end{vmatrix}$$

BASIC TOOL: ORIENTATION TESTS

2D application: Relative position point-circle

Observation

In order to compute the determinant of the previous corollary, it is convenient to do the calculations in terms of the differences between the values of the coordinates of the points involved, and to avoid making calculations (specially, products) in terms of the coordinate values, if possible:

$$\begin{vmatrix} x_1 & x_2 & x_1^2 + x_2^2 & 1 \\ a_1 & a_2 & a_1^2 + a_2^2 & 1 \\ b_1 & b_2 & b_1^2 + b_2^2 & 1 \\ c_1 & c_2 & c_1^2 + c_2^2 & 1 \end{vmatrix} = \begin{vmatrix} b_1 - a_1 & b_2 - a_2 & (b_1 - a_1)(b_1 + a_1) + (b_2 - a_2)(b_2 + a_2) \\ c_1 - a_1 & c_2 - a_2 & (c_1 - a_1)(c_1 + a_1) + (c_2 - a_2)(c_2 + a_2) \\ x_1 - a_1 & x_2 - a_2 & (x_1 - a_1)(x_1 + a_1) + (x_2 - a_2)(x_2 + a_2) \end{vmatrix}$$

Topic of lab 3

FURTHER READING

J. O'Rourke

Computational Geometry in C

Cambridge University Press, 1994 (2nd ed. 1998), pp. 17-35.

F. P. Preparata and M. I. Shamos

Computational Geometry: An Introduction

Springer-Verlag, 1985, pp. 36-45.